Bitcoin Price Prediction Using Machine Learning

Project Report

1. Project Overview

This project explores the use of machine learning algorithms to predict Bitcoin price movements using historical market data. The aim is to compare different regression models and identify a model that provides reliable predictions while remaining interpretable. The work follows a standard machine learning workflow including data preprocessing, feature engineering, model training, evaluation, and comparison.

2. Objectives

• Collect historical Bitcoin price data.
• Clean and preprocess the dataset.
• Train multiple machine learning models.
• Evaluate performance using RMSE and MAE.
• Compare models and identify the most suitable approach.

3. Dataset and Preprocessing

Historical Bitcoin market data containing price-related variables such as Open, High, Low, Close, and Volume was used. Missing values were handled, numerical features were normalized where appropriate, and the dataset was divided into training and testing subsets to evaluate generalization.

4. Methodology

Several regression algorithms can be evaluated, including Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting methods. Model performance is assessed on unseen test data using common regression metrics. Visualizations of actual versus predicted prices help understand model behavior.

5. Evaluation Metrics

Root Mean Squared Error (RMSE) measures the average prediction error while giving greater weight to large errors. Mean Absolute Error (MAE) measures the average magnitude of prediction errors. Lower RMSE and MAE indicate better predictive performance.

6. Example Model Comparison

Model	Strengths	Limitations
Linear Regression	Simple, interpretable	Limited for nonlinear patterns
Decision Tree	Easy to understand	Can overfit
Random Forest	Robust and accurate	Higher computational cost
Gradient Boosting	High predictive performance	Requires tuning

7. Results and Discussion

The purpose of this project is to demonstrate a structured machine learning workflow rather than claim a universally best model. Model selection should be based on evaluation results obtained on the specific dataset. Performance should be interpreted alongside model complexity, interpretability, and generalization.

8. Conclusion

This project demonstrates practical experience in data preprocessing, feature engineering, regression modeling, model evaluation, and comparative analysis. It highlights the importance of selecting models using objective evaluation metrics rather than relying solely on predictive accuracy. Future work could investigate deep learning models, additional financial indicators, and time-series forecasting methods such as LSTM architectures.

Skills Demonstrated

Python • Pandas • NumPy • Scikit-learn • Machine Learning • Regression • Data Visualization • Model Evaluation • RMSE • MAE • Jupyter Notebook